Yifan Yang

Research Assistant | M.Sc. in Artificial Intelligence and Robotics

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Summary

Research assistant advised by Prof. Pinjia He and M.Sc. student working at the intersection of AI for Software Engineering and AIOps, with a focus on microservice diagnostics, observability, fault injection, and LLM-based RCA agents.

Education

M.Sc. **The Chinese University of Hong Kong, Shenzhen** Sep 2024 – Present
Artificial Intelligence and Robotics Shenzhen
- School of Data Science.

B.Mgt. **Shandong University** Sep 2019 – Jun 2023
Industrial Engineering Jinan
- Training in systems thinking, optimization, and engineering practice.

Experience

Research Assistant Aug 2024 – present
The Chinese University of Hong Kong, Shenzhen Shenzhen
- Work under the supervision of Prof. Pinjia He at the School of Data Science.
- Conduct research in AI for Software Engineering and AIOps, with an emphasis on microservice diagnostics, root cause analysis, observability, and fault-propagation-aware evaluation.
- Contribute to end-to-end research workflows across multiple projects, including system implementation, telemetry and data collection, fault injection, system integration, baseline reproduction, and performance evaluation.
- Help maintain lab computing and networking infrastructure, including CPU clusters, GPU resources, Kubernetes environments, PVE virtualization, and OpenWrt-based networking.
- Designed fault-injection campaigns, improved observability integration across microservice systems, contributed to dataset organization, and wrote related sections for OpenRCA 2.0 and RCA benchmark research.
- First author of Gleaner, published in Proc. ACM Softw. Eng. (ISSTA 2026).

Publications

Rethinking the Evaluation of Microservice RCA with a Fault Propagation-Aware Benchmark June 2026
Aoyang Fang, Songhan Zhang, **Yifan Yang**, Haotong Wu, Junjielong Xu, Xuyang Wang, Rui Wang, Manyi Wang, Qisheng Lu, Pinjia He*
Proc. ACM Softw. Eng., 3(FSE):1605-1627, 2026 [PDF][DOI]

OpenRCA 2.0: From Outcome Labels to Causal Process Supervision June 2026
Aoyang Fang, **Yifan Yang**, Jin'ao Shang, Qisheng Lu, Junjielung Xu, Rui Wang, Songhan Zhang, Yuzhong Zhang, Boxi Yu, Pinjia He*
arXiv preprint [PDF]

Beyond Fault Localization: A Trajectory-Level Study of LLM Agents for Microservice Root Cause Analysis Aug 2026
Qisheng Lu, Aoyang Fang, Junjielong Xu, Jin'ao Shang, Songhan Zhang, **Yifan Yang**, Xiaochuan Yan, Pinjia He*
arXiv preprint [PDF]

Gleaner: A Semantically-Rich and Efficient Online Sampler for Microservice Diagnostics Oct 2026
Yifan Yang, Aoyang Fang, Songhan Zhang, Pinjia He*
Proc. ACM Softw. Eng., 3(ISSTA):3433-3453, 2026 [PDF][DOI][code]

DynaCausal: Dynamic Causality-Aware Root Cause Analysis for Distributed Microservices Oct 2025
Songhan Zhang, Aoyang Fang, **Yifan Yang**, Ruiyi Cheng, Xiaoying Tang, Pinjia He*
arXiv preprint [PDF]

Awards

Outstanding Academic Scholarship — CUHK-Shenzhen MAIR, 13 October 2026.

Service

- Artifact Evaluation Program Committee, ASPLOS 2027
- Artifact Evaluation Program Committee, ACM SOSP 2026
- Artifact Evaluation Program Committee, ACM CCS 2026
- Artifact Evaluation Program Committee, NDSS 2027

Projects

Gleaner — Diagnosis-oriented online trace sampling for microservice observability.
[GitHub]

chaos-experiment — Fault injection tooling for controlled microservice RCA evaluation.
[GitHub]

Aegis — Experiment tooling and systems engineering workspace for AIOps research.
[GitHub]

Skills

Research Areas: Intelligent Software Engineering (AI4SE), AIOps, intelligent observability, LLM-based and multi-modal RCA, fault injection benchmarks

Systems: Linux, Docker, Kubernetes, Helm, Skaffold, OpenTelemetry, Jaeger, PVE, OpenWrt

AI-assisted Development: Codex, Claude Code, Git, LaTeX, research automation workflows

ML and LLM: PyTorch, vLLM, LLM fine-tuning, inference optimization

Languages: Chinese (Native), English (IELTS 6.5)